

Hypothesis Testing

UC Berkeley, MIDS w203

Statistics for Data Science

UC Berkeley, School of Information

Speaker notes

No notes on this slide.

Introducing the Two-Sample t-Test

Motivating the Two-Sample t-Test

Examples:

- Are customers who get a birthday gift less likely to leave than those who don't?
- Do patients who take Vitamin W get over the flu faster than patients who don't?
- Do democracies or autocracies start more wars?

Speaker notes

- The two-sample t-test is one of the most common procedures in inferential statistics. What is it for? As a data scientist you will often face a question about two groups - group A and group B. Someone will ask you if one is greater or less than the other.
- Questions like this are everywhere. In business, you have A-B testing.
- In medicine, you may want to test a group of patients taking a vitamin against a control group
- In political science, you may want to test whether democracies or autocracies start more wars.
- What do these questions have in common? They all have two groups, and in all cases we're measuring the outcome on a metric scale. So it makes sense to ask about the expectation of each population - are the expectations different?

Example Scenario

Is this evidence that San Franciscans have more black outfits than New Yorkers *in expectation* ?

Speaker notes

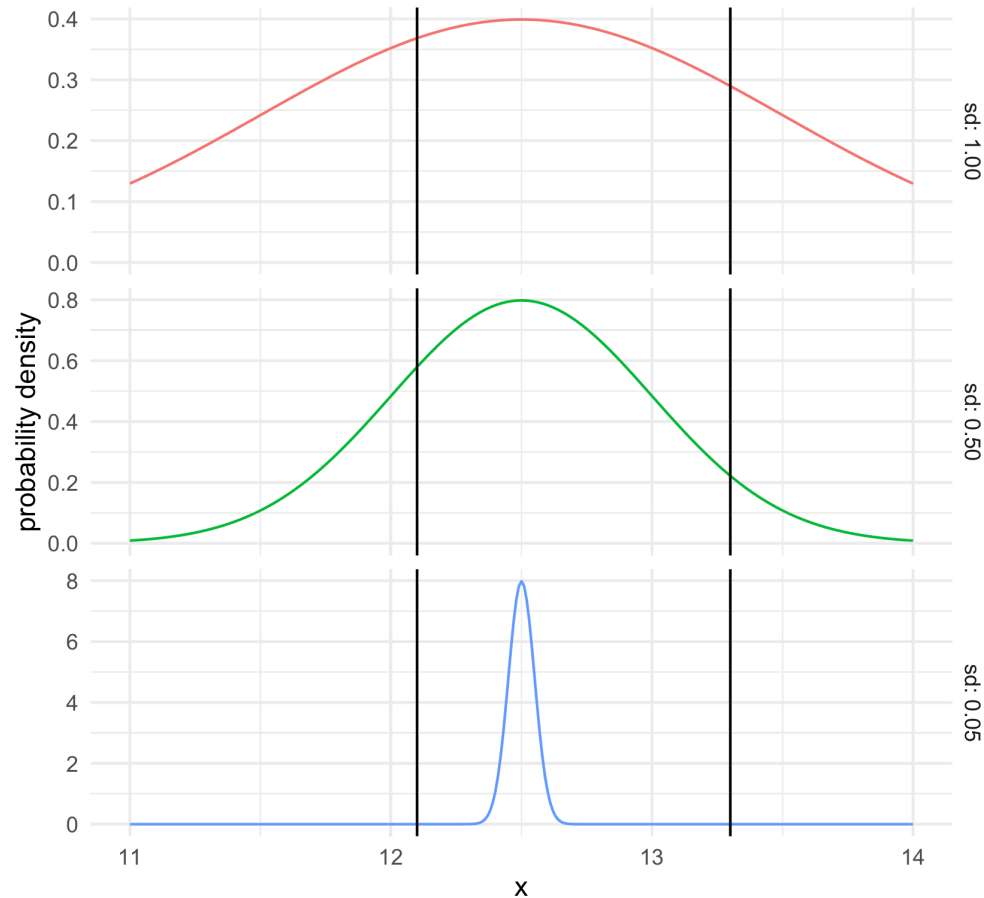
- Here's a specific example, with data.
- write on slide: $13.3 - 12.1 = 1.2$ outfits
- Is this a big difference? is it just noise?
- To be more precise, is this evidence that SF'ans have more black outfits than NYers in expectation?
- To analyze this question, we need a model...

Two-Sample Model Framework

Speaker notes

- A model is a representation of the world built of RVs. for a two-sample t-test, the model begins like this.
- Let
- Let's assume that the X 's are iid with mean
- We'll need more assumptions, but this is a start.
- The null is that our two means are the same.
- Our usual alternative is that the means are different.
- It is also possible to run a one-sided test in special circumstances. alt is one mean is greater than other.
- Given this model, how plausible is the null hypothesis?

Comparing Population Means



Speaker notes

- One important piece of information is how much does the number of black outfits vary in the population.
- Here's a picture with three different distributions for our sample averages.
- The top distribution has very high standard deviation. You can see that it doesn't seem unusual to get two sample averages that are 1.2 outfits apart.
- The middle distribution has medium deviation. Now it starts to look a little more surprising that our two sample averages were 1.2 apart.
- The bottom distribution has low standard deviation. Now it seems quite unlikely to get two sample averages 1.2 apart. The data seems inconsistent with this hypothesis.

Comparing Population Means

Speaker notes

- Putting these ideas together, we can make a statistic by taking the difference in sample averages, and dividing by an estimate of standard deviation - that's exactly the idea behind the two- sample t-test

The Two-Sample z-Test

Two-Sample -Test

Speaker notes

- Before we tackle the two-sample t-test, begin with the simpler 2 samp. z test. It's unrealistic, but build intuition.
- Same assumptions as before, but add the assumption of equal var
- How do we create a test statistic?
- What is the distribution of
- $V[\bar{X} - \bar{Y}] = V[\bar{X}] + V[\bar{Y}] = \sigma_X^2/n_X + \sigma_Y^2/n_Y$
- $\bar{X} - \bar{Y} \sim N(0, \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}})$
- We can standardize by dividing.
- Here's the standard normal. We can plot our statistic on this curve, to get a p-value or see if it's in a rejection region.

Let $X_1, \dots, X_{50}$ rep. New Yorkers. Assume iid, mean μ_X .

Let

$Y_1, \dots, Y_{50}$ rep. San Franciscans. Assume iid, mean μ_Y .

Assume

$$\sigma_X = \sigma_Y = 2.$$

Speaker notes

- Here's an example with actual data.
- $z = \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} = \frac{12.1 - 13.3}{\sqrt{\frac{4}{50} + \frac{4}{50}}} = \frac{-1.2}{\sqrt{\frac{4}{25}}} = -3$
- $z < -1.96$
- We can also compute p-value
 $2 * \text{pnorm}(-3) = .0027$

General Procedure for Testing

Three steps:

1. Specify model, null hypothesis, rejection criterion
2. Calculate statistic
3. Plot statistic on the null distribution to get the p value.

Speaker notes

- From this example, I hope you see that the two-sample z-test is not all that different from the one-sample tests we did earlier.
- No matter what the test is, there's a general pattern that you follow...

The Two-Sample t-Test

Types of Two-Sample t-Tests

1. Student's t-Test

2. Welch's t-Test

Speaker notes

- There are actually two versions of the two-sample t test you should be aware of.
- Student's t-test is the original t-test. it's simpler, but requires strong assumptions
- Welch's t-test is more general, and this is really the modern t-test.

The Two-Sample z-Test

$$Z = \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}}} - \text{Problem : we don't know } \sigma_X \text{ or } \sigma_Y$$

Speaker notes

- To explain these tests, let's remember our equation for the z test
- The big problem with this test is we're using the standard deviations
- We're going to have to estimate these in some way.

Student's t-Test

- Estimate a single “pooled” standard deviation, s .

- Substitute s for both s_X and s_Y .
$$t = \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{s^2}{n_X} + \frac{s^2}{n_Y}}}$$

Speaker notes

- Here's what Student did..
- Student's idea is to create a single estimate for standard deviation - let's call it
- I'm not including the equation for
- You plug
- Why would you do this? Well, it turns out that Student got a really clean, elegant result
- Theorem: this statistic follows a t dist. with

- **One sample t-test** - Model has one parameter (the mean)
- Given the sample mean, and $n - 1$ observations, can compute the last one.
- $df = n - 1$
- **Student's two-sample t-test** - Model has two parameters (μ_X and μ_Y)
- Given the sample means, $n_X - 1$ observations for X and $n_Y - 1$ observations for Y , can compute the rest
- $df = n_X + n_Y - 2$

Speaker notes

- Why is there a -2 in degrees of freedom?
- There's a pattern. Think of degrees of freedom in this way. (read def)
- For the one-sample test, our model had one parameter - the mean. it's the mean we wanted to test.
- If I gave you the sample mean, and $n-1$ rows of data, you could compute the last row.
- There's no more information in the last row of data. it's already locked in.
- so df is n minus 1 for the estimated parameter.
- For the two-sample test, we now have a model with two parameters...

Student's t-Test Summary

- Tests if mean of X equals mean of Y .
- Uses a pooled estimate for standard deviation.
- **Major disadvantage:** Only valid if $\sigma_X = \sigma_Y$.

Speaker notes

- Let's take a moment to summarize Student's t-Test.
- We use the t-Test to see if
- Student used a pooled estimate for standard deviation.
- But there's one problematic assumption: the test is only valid if

Welch's t-Test

- Compute two sample standard deviations: s_X and s_Y .
- Substitute s_X for σ_X and s_Y for σ_Y .

$$t = \frac{\overline{X} - \overline{Y}}{\sqrt{\frac{s_X^2}{n_X} + \frac{s_Y^2}{n_Y}}}$$

Speaker notes

- Let's move on to Welch's test.
- Unlike Student, Welch computed two different sample standard deviations.
- Then substitute in
- Unfortunately, this method just isn't as clean mathematically.
- You do get a similar theorem, but (1) the distribution is only approximately t. and (2) there's this complicated equation for degrees of freedom. the degrees of freedom you get are usually not an integer.

Choosing a Two-Sample t-Test

Some authors recommend a two step process:

1. Use Levine's test for equal variances ($H_0 : \sigma_X = \sigma_Y$)
2. If non-significant, proceed with Student's t-Test

Our advice: always use Welch's t-Test

- Power is almost as high as for Student's test
- We never know for sure if variances are equal
- This is the default in most statistical software

Speaker notes

- Why is this our advice? 3 reasons
- First, the only reason you'd ever want more assumptions is to get more power. It's true that Student's test is more powerful - but studies have shown that the difference is too small to worry about
- Second, you never know if the variances are equal - you can use a test, but a test is just a test - it's not proof. and the result really depends on the sample size. for small samples in particular, it's really hard to know.
- Finally, this is the default setting in most software. not really a reason, but it is convenient.

Welch's Two-Sample t-Test Assumptions

- **Metric Scale:** $X_1, X_2, \dots, X_{n_X}$ and $Y_1, Y_2, \dots, Y_{n_Y}$ are random variables measured on a metric scale.
- **Independence:** X 's are iid, Y 's are iid, and X 's and Y 's are mutually independent.
- **Normality:** The distribution of the X 's is normal and the distribution of the Y 's is normal - The CLT guarantees normality for large samples
- Main concern is strong skewness with a small sample

- Let's summarize the assumptions in a more user-friendly form.

Practical Significance of the T-Test

Practical Significance for the T-Test

After using a t-test to assess statistical significance, it is important to assess practical significance.

Three common effect size measures: 1. Difference in means 2. Cohen's d 3. Correlation r

Difference in Means

Difference in means

$$[\{A\} - \{B\}]$$

- Answers the question `` *How different are these groups?*’’
- Often makes great headlines and is a good choice if units are familiar
- But lacks context in its calculation
- People who eat chocolate live 1.5 years longer than those who do not eat chocolate

Cohen's d

Cohen's d

Cohen's d is a measure of difference of means standardized by the variance in the data.

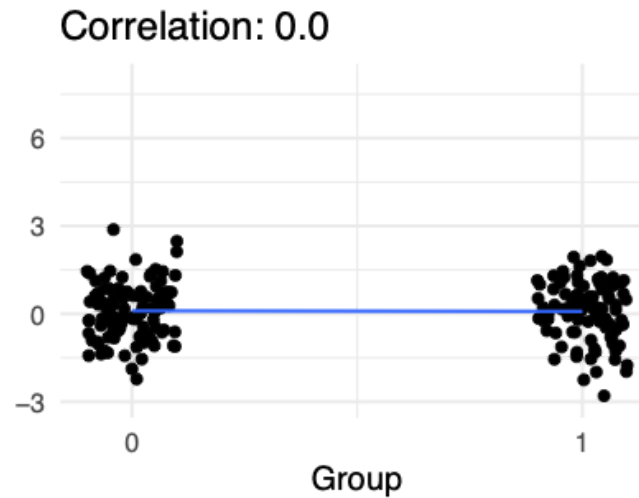
[] Where s is a pooled standard deviation: $\sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2}}$

- Answers the question `` *How many standard deviations apart are the groups?* ''
- The difference in sarcasm score between frequentists and Bayesians is $d = 0.54$ standard deviations.

Correlation

i Biserial Correlation

Correlation answers the question ``How strong is the relationship between group identity and the outcome?’’ [=]



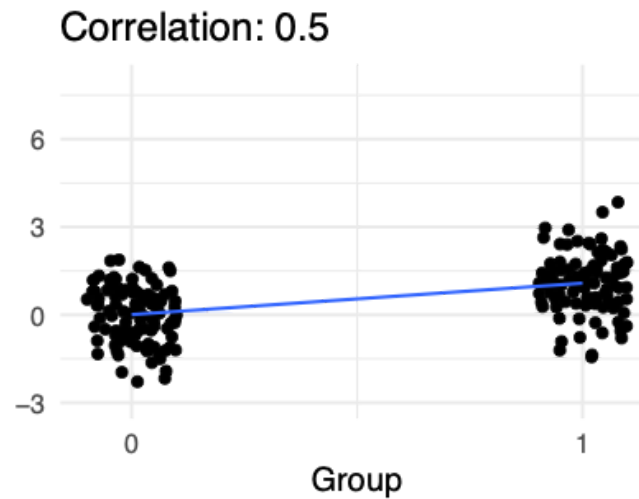
Speaker notes

- Notice the similarity in the form between Cohen's d and correlation – Cohen's d divides by the pooled standard deviation; correlation divides by the product of two group standard deviations.

Correlation

i Biserial Correlation

Correlation answers the question ``How strong is the relationship between group identity and the outcome?’’ [=]



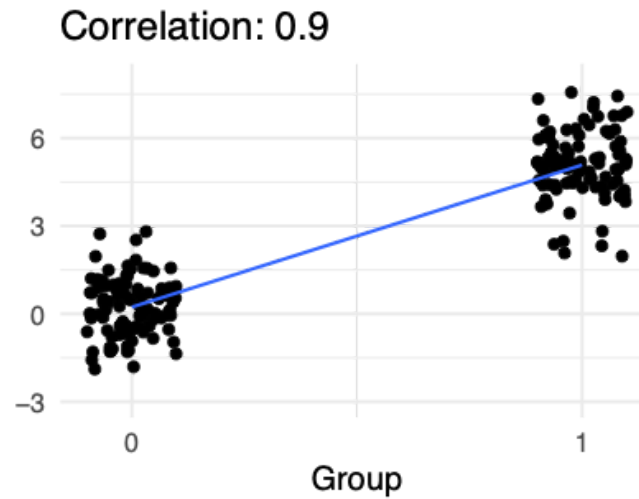
Speaker notes

No notes on this slide.

Correlation

i Biserial Correlation

Correlation answers the question ``How strong is the relationship between group identity and the outcome?’’ [=]



Speaker notes

No notes on this slide.

Practical Significance is about Context

- How strong is the same relationship between *different* groups?
- How strong is a *different* relationship between the same group?
- What is the underlying dispersion in the data?
- What is a meaningful anchor or reference point that you can use for context?

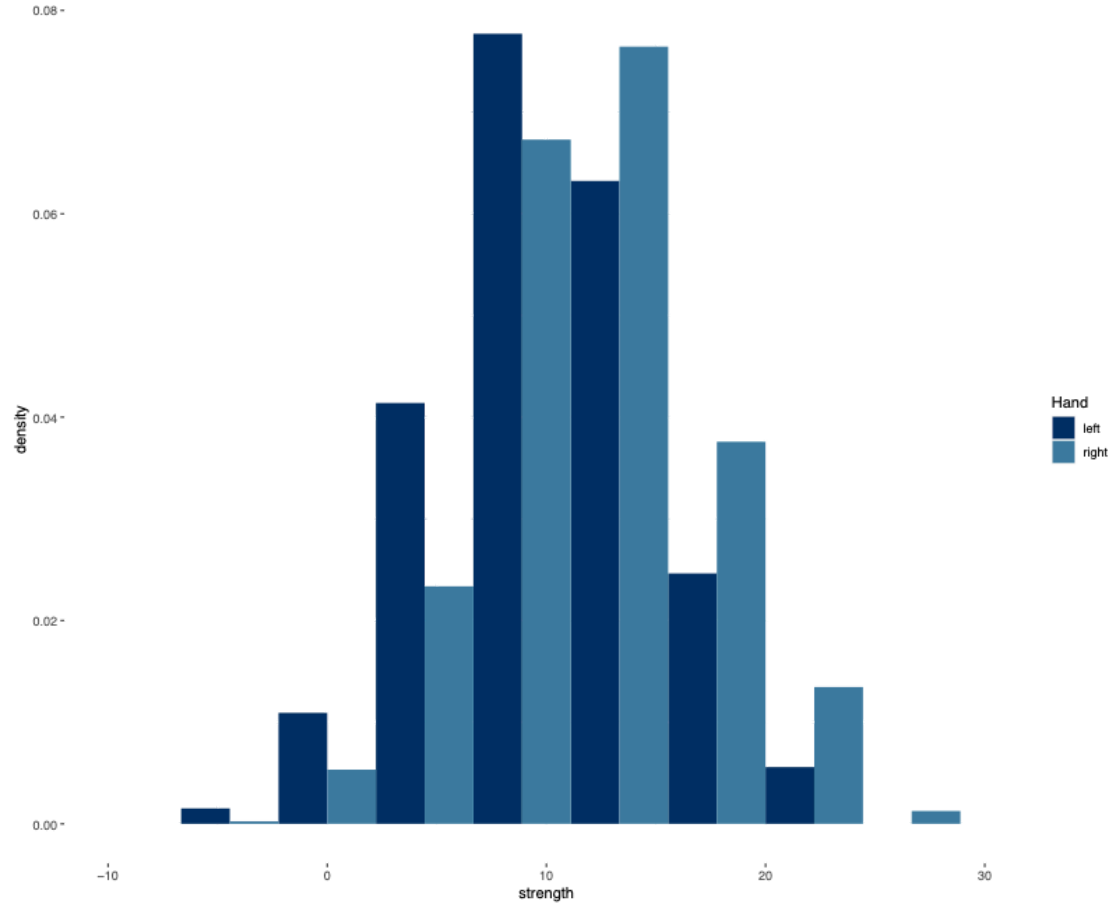
The Paired t-Test

Paired t-Test

Speaker notes

No notes on this slide.

Paired t-test

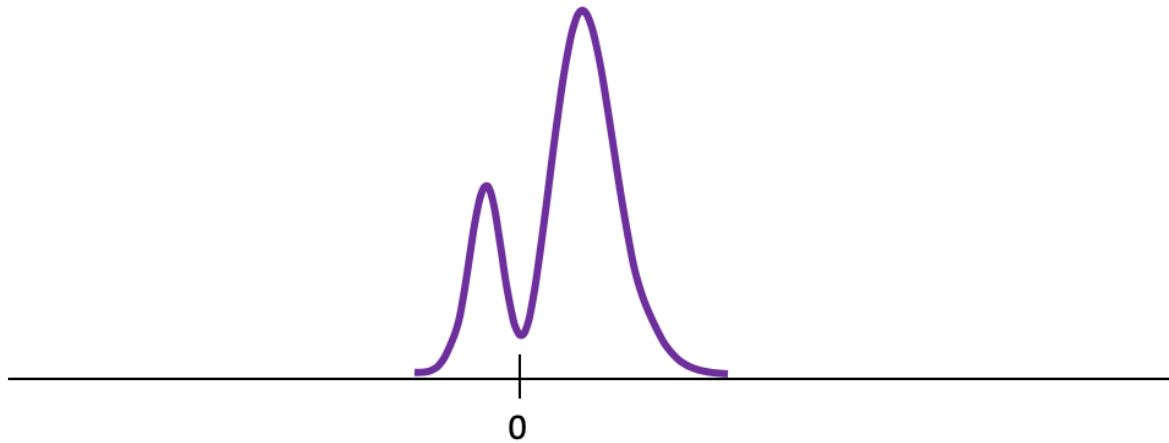


Speaker notes

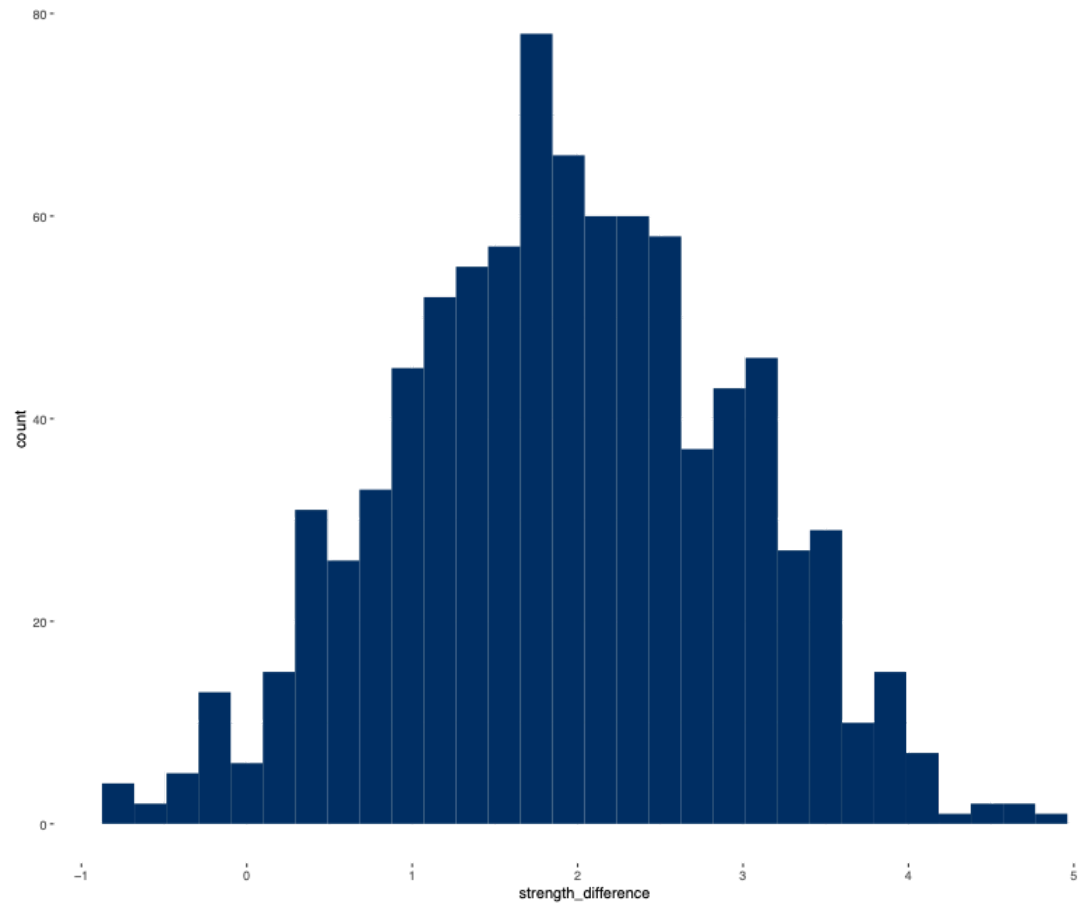
No notes on this slide.

Paired t-Test

- **Idea:** For any *particular* subject i , the difference between right-hand strength and left-hand strength, $R_i - L_i$, will usually be small.
- Within-person variation is small.



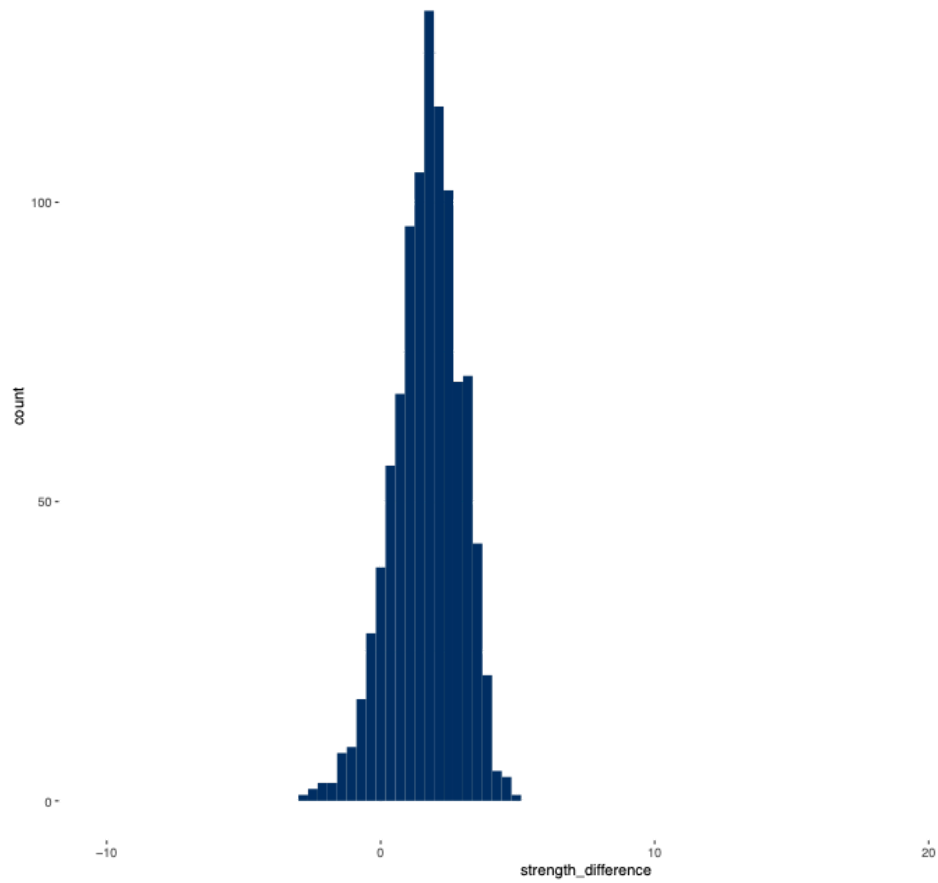
Paired t-Test



Speaker notes

No notes on this slide.

Paired t-Test



Speaker notes

No notes on this slide.

Paired t-Test

Paired t-test

A `_paired t-test_`, sometimes called a `_dependent t-test_`, builds an explicit dependency between data. Instead, perform a one-sample t-test with $H_0: \mathbb{E}[R_{\{i\}} - L_{\{i\}}] = 0$.

- This dependency must actually exist
- Cannot simply change the test

Unpaired vs. Paired t-Test

$$\text{Unpaired- } t = \frac{\overline{A-B}}{\sigma_{A\&B}}$$

$$\text{Paired- } t = \frac{\overline{A-B}}{\sigma_{(A-B)}}$$

Paired t-Test Assumptions

- A and B have a metric scale with the same units.
- There is a natural pairing between observations for A and for B . - pre-test and post-test for same individual
- response to two types of stimulus for same mouse
- responses for a pair of spouses
- Each pair (A_i, B_i) is drawn i.i.d.
- The distribution of $A - B$ is sufficiently normal given the sample size.

Introduction to Non-parametric Tests

Non-parametric Tests

- t -test is parametric, like all the tests we've seen so far -
Assumes the population comes from a parametric family of distributions
- Typically the normal curves
- It is not always possible to meet this assumption

Non-parametric Tests (cont.)

Large sample - No Problem - central limit theorem tells us that the sampling distribution of the mean will be approximately normal, so t -tests are valid - Parametric tests are generally valid for large samples

Non-parametric Tests (cont.)

Small sample - t -test is fairly robust to deviations from normality, but you should look at your distribution and see how non-normal it is - Suppose you have a small sample and you suspect you have a major deviation from normality - You might be able to transform the variable to make it more normal, but that can alter the meaning and make results harder to interpret

Non-parametric Test Details

- Non-parametric tests can be also called **distribution-free tests** - Still involve assumptions, but they are less restrictive than those of parametric tests
- Many tests work on principle of ranking data - List the scores from lowest to highest – each score gets a rank, so higher scores have higher ranks
- Only consider ranks instead of looking at the metric value of the variable
- Use the order of variables to construct statistics that we can use to test hypotheses

Non-parametric Test Details (cont.)

Advantages- Population distribution doesn't have to be normal - Easier to justify a rank-based test

Disadvantages- We throw out metric information - Rule of thumb: if you throw away information, you lose statistical power

Rank-Based Tests for Ordinal Variables

- Rank-based tests are especially useful when we have an ordinal variable - eg. a Likert variable such as “how do you feel about a presidential campaign?”
- Neutral, support, strongly support, etc.
- It is hard to argue that the difference between neutral and support is the same as the difference between support and strongly support

Love Tester Example



Do you trust that the difference between harmless and mild is the same as the difference between burning and passionate?

Rank-Based Tests for Ordinal Variables (cont.)

If you run a t -test in these cases, you impose a linear structure on your variable, treating it as metric

- This method may or may not be reasonable
- If you use a rank-based test that is okay—you are asking whether one group tends to rank below or above another
- The ranks are still meaningful

Conclusion

- There are some situations in which you should consider non-parametric tests
- Coxe is going to tell you more about the specifics

Wilcoxon Rank-sum Test for Independent Groups

Parametric and Non-parametric Tests for Comparing Only Two Groups

Comparing Two Independent Conditions: Wilcoxon Rank-Sum Test

- Data are ranked from lowest to highest across groups
- This provides **potential rank** scores
- If the same score occurs more than once then all scores of the same value receive the average of the potential ranks for those scores
- This gives us the **final rank** scores

Comparing Two Independent Conditions: Wilcoxon Rank-Sum Test

Calculating the Wilcoxon Rand-Sum Test

- After assigning final ranks, add up all the final ranks for each of the two groups
- Subtract the mean rank for a group of the same size as our groups - Otherwise, larger groups would always have larger values
- For example, the mean group for a group of four = $1 + 2 + 3 + 4 = 10$
- Our final calculation is therefore: - $W = \text{sum of ranks} - \text{mean rank}$

Calculating the Wilcoxon Rank-Sum Test (cont.)

- Group A: $W = \text{sum of ranks (17.5)} - \text{mean rank (10)} = 7.5$

Interpretation of the Wilcoxon Rank-Sum Test

- You can also do a one-directional test if you hypothesize that one particular group will have higher ranks than the other
- Always two values for W (one for each group)
- Lowest score for W is typically used as the test statistic

Interpretation of the Wilcoxon Rank-Sum Test (cont.)

- For small sample sizes ($N < 40$), R calculates the p value with the Monte Carlo methods - ie. simulated data are used to estimate the statistic
- For larger samples, R calculates the p value with a normal approximation method - Assumes that the sampling distribution of the W statistic is normal, not the data
- Normal approximation method helpful because it calculates a z statistic in the process of calculating the p value

Effect Size for the Wilcoxon Rank-Sum Test

Speaker notes

No notes on this slide.